ELEC 221 Lecture 15
Analysis of CT systems based on differential equations

Thursday 29 October 2026

Announcements

- Midterm Monday during tutorial section
- Office hours tomorrow at 10am (instead of 12pm)

Last time: clarification

Correction: solution to quiz question.

$$\begin{aligned} H(j\omega) &= \frac{1}{2+j\omega}\frac{1}{5+j\omega} \\ x(t) &= \cos(3t) \end{aligned}$$

Last time

We analyzed the impulse and step response of an LTI system described by a first-order differential equation.

$$T\frac{dy(t)}{dt} + y(t) = x(t),$$

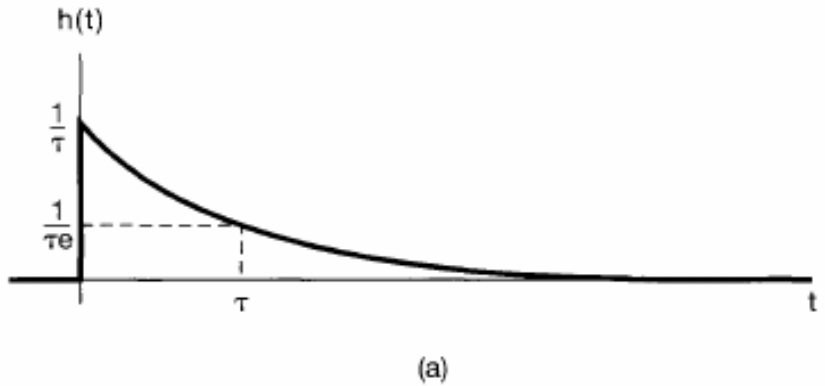


Image: Oppenheim Fig. 6.19

Last time

We derived the behaviour of the Fourier transform under differentiation and integration:

From these, we determined

Today

Learning outcomes:

- compute the Fourier spectrum of CT LTI systems described by differential equations
- characterize the oscillatory behaviour of CT systems described by second-order differential equations

Example: Fourier transform properties and differentiation

We can take advantage of differentiation and integration properties to simplify computations.

Suppose

What is the Fourier transform of

Two properties to take advantage of here:

Example: Fourier transform properties and differentiation

$$z(t) = \frac{d^2}{dt^2} x(t-1)$$

First, consider: $p(t) = x(t-1)$:

Fourier transforms and systems described by differential equations

Recall the LTI system described by a first-order ODE:

LTI systems can also be described by higher-order ODEs:

We will derive a general expression for the frequency response of such systems.

Fourier transforms and systems described by differential equations

$$\sum_{k=0}^{N} \alpha_k \mathcal{F}\left(\frac{d^k y(t)}{dt^k}\right) = \sum_{k=0}^{M} \beta_k \mathcal{F}\left(\frac{d^k x(t)}{dt^k}\right)$$

Fourier transforms and systems described by differential equations

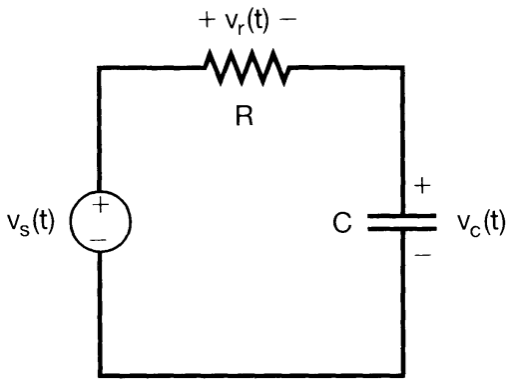
$$Y(j\omega)\sum_{k=0}^{N}\alpha_k(j\omega)^k = X(j\omega)\sum_{k=0}^{M}\beta_k(j\omega)^k$$

Final property:

This representation allows us to write down frequency response of systems described by ODEs **by inspection**! (and vice versa)

Exercise: frequency response of systems described by ODEs

What are the **impulse response** and **frequency response** of our RC circuit filter?



Exercise: frequency response of systems described by ODEs

What are the **impulse response** and **frequency response** of the system described by

Start with frequency response:

Example: frequency response of systems described by ODEs

We can now leverage this to determine the impulse response:

$$H(j\omega) = \frac{3(j\omega)^2 + 1}{(j\omega)^3 - 4j\omega}$$

Use partial fractions:

Example: frequency response of systems described by ODEs

Details are left as an exercise:

To get the impulse response, we can take the inverse Fourier transform, and leverage linearity:

Example: frequency response of systems described by ODEs

$$h(t) = -\frac{1}{4}\mathcal{F}^{-1}\left(\frac{1}{j\omega}\right) + \frac{13}{8}\mathcal{F}^{-1}\left(\frac{1}{j\omega+2}\right) + \frac{13}{8}\mathcal{F}^{-1}\left(\frac{1}{j\omega-2}\right)$$

Check Table 4.2 - two expressions to leverage:

So we have:

Second-order systems

Consider a system described by the ODE

$$\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 x(t)$$

Exercise: what is the frequency response?

Second-order systems

Let's explore this in a little more detail and compute the impulse and step response of this system.

where

Three cases to consider:

- $\zeta = 1$
- $\zeta > 1$
- $\zeta < 1$

Second-order systems

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}, \quad c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case: $\zeta = 1$.

Use handy table of Fourier transform pairs to find

Second-order systems

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}, \quad c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case: $\zeta \neq 1$. Do a partial fraction expansion:

Use table of Fourier transform pairs to find

Second-order systems

$$h(t) = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(e^{c_+ t} - e^{c_- t}\right) u(t), \qquad c_\pm = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

Exercise: derive the step response,

$$s(t) = \left[1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(\frac{e^{c_+ t}}{c_+} - \frac{e^{c_- t}}{c_-}\right)\right] u(t)$$

The form of the exponential depends whether $\zeta > 1$ or $\zeta < 1$

- $\zeta < 1$: $c_\pm$ are imaginary; complex exponentials, so the response will oscillate!
- $\zeta > 1$: $c_\pm$ real and negative; decaying exponentials

Let's go plot these.

For next time

Content:

- Discrete-time FT
- New chapters: sampling; moodulation

Action items:

1. Prepare for midterm

Recommended reading:

- From today's class: Oppenheim 4.3.4, 6.5, 6.7
- Suggested problems: 4.33-4.36, 6.5, 6.9, 6.15, 6.18, 6.26, 6.27
- For next class: Oppenheim Ch. 5